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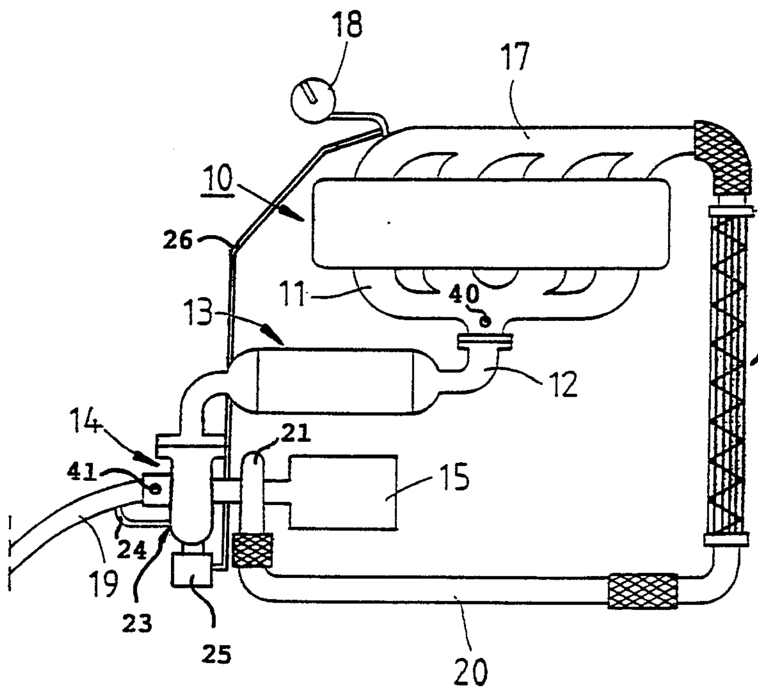

(57) Abstract

The object of the invention is a combustion engine apparatus comprising a diesel-powered Otto-cycle engine (10), an exhaust supercharger (14) and a catalytic converter (13). According to the invention the passive catalytic converter (13) is mounted between the combustion engine (10) and the exhaust supercharger (14), that is, the exhaust gases are led to the catalytic converter directly from the engine, before the supercharger.

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COMBUSTION ENGINE APPARATUS

The object of the invention is a combustion engine apparatus comprising a diesel-powered Otto-cycle engine, an exhaust supercharger and a catalytic converter.

Today, catalytic converters are used widely in combustion engines to purify engine exhaust gases of impurities. Combustion engine exhaust gases normally contain, for example, extremely toxic carbon monoxide, hydrocarbons, and aldehydes which irritate the mucous membranes of the mouth, throat and nose.

There are several types of catalytic converters. However, they all have in common the functioning principle according to which the purifying effect of the catalytic converter is based on the action of a noble metal, usually platinum, as a catalytic element. Due to the effect of the catalytic converter, harmful gases react with oxygen, converting them into less harmful compounds. In this way, for example, the extremely hazardous carbon monoxide can be oxidized to form carbon dioxide, and the hydrocarbons oxidize to form carbon dioxide and water.

A prerequisite for the efficient functioning of the catalytic converter is, however, that the temperature of the exhaust gases passing through the catalytic converter is sufficiently high. The purifying action of the catalytic converter does not usually start until the temperature of the exhaust gases exceeds 200°C and the operation of the catalytic converter is usually efficient at a temperature of 300°C or higher.

The problem with diesel engines is that, when idling, the temperature of the exhaust gases is only about 120-140°C, and thus at these low temperatures the catalytic converter does not function at all. Even under a partial load, however, the temperature of the diesel engine does rise to the catalytic converter's operating range, that is, to a temperature of about 300-350°C. In the case of a conventionally aspirated engine the catalytic converter actually functions well in such a case, but the exhaust supercharger incorporated in the engine causes problems.

Today, turbochargers are used in an increasing number of diesel engines to increase engine output and to improve the fuel economy of the engine. One drawback is that, in an exhaust supercharger, the passing of the exhaust gases through a turbine based on conventional pulsating flow causes the exhaust gases to cool. The temperature of the gases falls by about 150°C as they pass through the turbine, because the exhaust gases release kinetic and thermal energy for rotating the turbine. As a result, the temperature of the exhaust gases of a diesel engine running under partial load is only 150-200°C after the exhaust supercharger turbine. This temperature is below the operating temperature range of the catalytic converter, and thus the catalytic converter no longer purifies the exhaust gases.

US-patent publication 4 449 370 discloses a turbine based on conventional pulsating flow, which is used in connection with an aircraft engine. In this turbine, the exhaust gases are led through an apparatus named a catalytic combustor. To render the functioning of the catalytic combustor possible, an after-burner with a system for injecting additional fuel has been added to the catalytic combustor system because, as a result of the decrease in the kinetic energy of the exhaust gases, the catalytic combustor would not otherwise perform its intended task due to the low prevailing temperature. In different solutions, it is known to use the supercharger compressor shaft for rotating other devices such as a hydraulic pump, which means that obtaining the necessary power output requires a separate, additional fuel supply.

The object of the present invention is to eliminate the foregoing problem and to achieve a novel combustion engine apparatus, in which the catalytic converter functions efficiently, also when provided with an exhaust supercharger. It is characteristic of the invention that a passive catalytic converter is mounted between the combustion engine and the exhaust supercharger, that is, the exhaust gases are led to the catalytic converter straight from the engine, before the supercharger.

The advantage of the invention is that the exhaust gases are led to the catalytic converter directly from the engine, before the supercharger, in which case the temperature of the exhaust gases is in most situations sufficiently high for the effective functioning of the catalytic converter.

However, from the foregoing arrangement follows the drawback that, when passing through the catalytic converter, the exhaust gases lose some of their kinetic energy. And it is the kinetic energy of the exhaust gases which rotates the turbine of the exhaust supercharger. As a result of the decrease in the kinetic energy of the exhaust gases, the power output of the exhaust supercharger turbine decreases, and correspondingly, the power output of the supercharger connected to the turbine also decreases. Therefore, it is not possible to use a turbine based on conventional pulsating flow when the exhaust gases are led first through a catalytic converter.

According to the present invention, the problem relating to the decrease in the kinetic energy of the exhaust gases has been solved by using an exhaust supercharger turbine operating by means of laminar flow after the catalytic converter, the said turbine being connected to a passive catalytic converter which functions without a separate burner. The efficiency of the turbine is increased through the catalytic reaction taking place in the catalytic converter, the said reaction raising the temperature of the gases produced in the oxidization reaction, and at the same time alleviating the problem.

The function of the exhaust supercharger is to increase the pressure of the intake air supplied to the engine, thus providing the engine cylinders with sufficient air at all times. A sufficient supply of oxygen is a prerequisite for complete combustion of the fuel supplied to the cylinders, which means that the amount of harmful impurities produced is as low as possible.

For example, when the engine is subjected to a load and the supercharging pressure of the exhaust supercharger is 0,65 bar, the temperature of the supercharged air is usually 120-160°C, depending on the type and size of supercharger. Due to the hot intake air supplied to the engine, the temperatures of combustion and of the discharging gases also rise. In this case the disadvantage is that the amount of harmful nitric oxides increases considerably at temperatures exceeding 700°C.

To lower the combustion temperatures and to decrease the amount of nitric oxides, intercooling of the supercharging air is applied. In the apparatus relating to the invention an intercooler is used, which, together with the intake conduits, causes a pressure loss amounting to at most 200 mmH_2O. After the intercooler, the temperature of the supercharging air is also only 20-30°C higher than that of the ambient air. The temperature of the exhaust gases thus falls by about 200°C.

Cooling the intake air also has other advantageous effects. The thermal load on the pistons is reduced in the same proportion and the viscosity of the lubricating oil remains higher. Lubricating oil cannot, therefore, pass so easily beyond the piston rings into the combustion chamber to increase the hydrocarbon impurities of the exhaust gases.

The invention is described in the following by means of examples with reference to the enclosed drawings which show diagrammatically the combustion engine apparatus, and graphically the measurement results obtained in testing the apparatus.

Figure 1 shows the combustion engine apparatus relating to the invention.
Figures 2, 3 and 4 show graphically the test results obtained from testing the apparatus.

Figure 1 shows how the exhaust gases of a supercharged Otto-cycle engine 10 operating with diesel fuel are led through the exhaust manifold 11 to the exhaust conduit, in which a temperature sensor 12 is located. Differing from conventional exhaust supercharger design, which is based on pulsating technique, in which the supercharger is mounted as close to the exhaust ports of the cylinders as possible, in the solution relating to the invention the catalytic converter 13 converts the flow of the exhaust gases into a laminar flow, which means that changes are required also in the turbine structure of the supercharger, to render it suitable for laminar flows. The function of the supercharger 14 is to act as an additional air feeding compressor without additional devices connected to the turbine shaft.

Immediately connected to the exhaust conduit is a passive, double-acting catalytic converter 13, which functions without a separate burner. The double-acting catalytic converter purifies the carbon monoxide, hydrocarbons (HC) and aldehydes. In diesel engines, nitric oxides (NO_x) can only be reduced by lowering the combustion temperature. In the solution shown in the figure, the exhaust gases are led further on after the catalytic converter 13 to the turbine of the exhaust supercharger 14, and further to the exhaust pipe 19. The exhaust supercharger 14 is provided with a waste gate 23, which is connected by means of a pipe unit

24 between the supercharger 14 and the exhaust pipe 19. Some of the exhaust gases are discharged through the waste gate 23, when the pressure in the supercharger 11 rises above the set maximum pressure. This prevents the turbine from breaking should the turbine be subjected to strain caused by excessive overpressure, and overheating.

The intake air of the engine 10 is led through the air filter 15 to the compressor 21, which is provided with a maximum supercharging pressure regulator 25. The supercharged air is led through piping 20 to the intercooler 16 and further on to the inlet manifold 17 of the engine 10. A sensor 18 and the maximum supercharged pressure regulating unit 25 are connected to the inlet manifold 17 by means of a pipe unit 26, the said regulating unit adjusting the maximum supercharging pressure from the compressor 21 to suit the engine 10. A measuring element 40 is fixed to the exhaust manifold 11, and a measuring element 41 is fixed at the exhaust discharge opening of the exhaust supercharger 14, for measuring the temperatures of the exhaust gases and for simultaneously determining the degree of purification.

Figures 2, 3 and 4 show graphically the test results from the testing of the combustion engine apparatus under normal driving conditions. The combustion engine exhaust gases normally contain, for example, extremely toxic carbon monoxide, hydrocarbons, and aldehydes which irritate the mucous membranes of the mouth, throat and nose. With modern engines the amount of carbon monoxide may vary from about 0.05% up to 0.5%. In catalytic converters, noble platinum set in balls, ceramics or sheet metal bent into a wavelike shape, as in the present embodiment, is used as a catalytic element. By means of a catalytic element fixed in as thin a layer as possible on as large an area as possible, carbon monoxide and oxygen are oxidized to give carbon dioxide, and the hydrocarbons and oxygen are converted into carbon dioxide and water. In order for the catalytic purification to be successful, a sufficiently high reaction temperature is required. When the fastest possible rise in temperature is achieved in the catalytic converter, the best possible purifying effect and less soot are obtained. The exhaust purifier 13 already converts at 300°C over 90% of the carbon monoxide, 70% of the hydrocarbons and 70% of the aldehydes into harmless substances. In the case of carbon monoxide the purification already starts at 190°C, while 40% of the hydrocarbons are purified at 200°C and 80% at 300°C, and 75-80% of the aldehydes are purified at 300°C.

Figure 2 shows the temperature curves of the exhaust gases of the combustion engine apparatus when the supercharging pressure of the vehicle's exhaust supercharger 14 compressor 21 is adjusted to 0.4 bar, which corresponds to idling of the engine 10. Curve 30 shows the temperature of the exhaust gases at measurement point 40, before the supercharger 14 and the catalytic converter 13. The average of the temperatures measured at the measurement point 40 is about 300°C. Curve 31 correspondingly shows the temperature, at the measurement point 41, of the gases passing into the exhaust manifold 19 after the supercharger 14. The average of the temperatures measured at the measurement point 41 is about 200°C. Since the catalytic converter 13 is mounted between the combustion engine 10 and the exhaust supercharger 14, that is, the exhaust gases are led to the catalytic converter directly from the engine, before the supercharger, the catalytic converter 13 reaches the temperature required for exhaust gas purification very rapidly after the starting and departure of the vehicle. Curves 30 and 31 show the temperature variations during driving, under varying loads. At point P the vehicle is running under a steady load, in which case the temperatures in the combustion engine apparatus also stabilize.

Figures 3 and 4 show correspondingly the temperature curves of the exhaust gases when the engine is under a load of 0.6 and 0.8 bar supercharging pressure. The exhaust gases heat up more rapidly and the temperature differences between the curves 30 and 31 are greater. The measurement conditions shown in figure 3 correspond to normal town driving, in which case the load on the engine is higher and the driving speed greater, thus also rendering the temperature of the catalytic converter 13 higher and the purification degree even more efficient. The average of the temperatures in the catalytic converter 13 is 430°C, and correspondingly, after the supercharger 330°C, the purification degree being almost 100% regarding all emissions. By means of the combustion engine apparatus good purification efficiency is achieved in urban traffic under as small a load as possible and within a short running period.

The conditions shown in figure 4 correspond to highway speeds, that is, loading of the engine 10 at about 100 km/h. The average temperatures at measurement points 40 and 41 are 600°C and 470°C, respectively.

When the exhaust emissions of a vehicle provided with the combustion engine apparatus relating to the invention were tested in simulated town driving, the values for the vehicle were below the limit values of E-regulation no. 15/04 for a commercial vehicle of a corresponding reference weight range by a considerable margin. When the engine 10 was loaded in a chassis dynamometer while the amount of smoke was measured, the values produced by the vehicle tested were lower than the permitted maximum values specified in E-regulation no. 24/03.

It is obvious to one skilled in the art that the different embodiments of the invention may vary within the scope of the claims presented below.

CLAIMS

1. A combustion engine apparatus comprising a diesel-powered Otto-cycle engine (10), an exhaust supercharger (14) and a catalytic converter (13), **characterized** in that a passive catalytic converter (13) is mounted between the combustion engine (10) and the exhaust supercharger (14), that is, the exhaust gases are led to the catalytic converter straight from the engine, before the supercharger.

2. A combustion engine apparatus as claimed in claim 1, **characterized** in that the turbine of the exhaust supercharger (14) is a turbine operating by laminar flow, the said turbine operating by means of the laminar flow of the exhaust gases after the catalytic converter (13).

3. A combustion engine apparatus as claimed in claim 1 or 2, **characterized** in that the intake air of the combustion engine (10) is led to the inlet manifold (17) through the type of intercooler (16) which, together with the intake conduits (20), causes a pressure loss amounting to at most 200 mmH_2O.

4. A combustion engine apparatus as claimed in claim 1, 2 or 3, **characterized** in that the intercooler (16) is dimensioned so that the temperature of the supercharging air is 20-30°C higher than that of the ambient air after the intercooler.

5. A combustion engine apparatus as claimed in any of the claims 1-4, **characterized** in that the combustion engine (10) is a diesel engine, to the exhaust manifold (11) of which is connected a passive catalytic converter (13) through which exhaust gases are led to the exhaust supercharger (14) turbine operating by means of laminar flow, and that the intake air of the engine is led from the supercharger to the inlet manifold (17) of the engine

through the type of intercooler (16) which, together with the intake conduits (20), causes a pressure loss amounting to at most 200 mmH_2O, so that the temperature of the supercharging air is 20-30°C higher than that of ambient air after the intercooler.

6. A combustion engine apparatus as claimed in any of the claims 1-5, **characterized** in that a measuring element (40) is fixed to the exhaust manifold (11), and at the exhaust discharge opening of the exhaust supercharger (14) is fixed a measuring element (41) for measuring the temperatures of the exhaust gases and for simultaneously determining the degree of purification.

7. A combustion engine apparatus as claimed in any of the claims 1-6, **characterized** in that the catalytic converter (13) mounted between the combustion engine (10) and the exhaust supercharger (14) reaches, immediately after starting the engine (10), preferably at 0.4 bar supercharging pressure, the temperature required for exhaust gas purification.

AMENDED CLAIMS

[received by the International Bureau on 12 October 1992 (12.10.92); original claim 1 amended; remaining claims unchanged (2 pages)]

1. A combustion engine apparatus comprising a diesel-powered Otto-cycle combustion engine (10), an exhaust supercharger (14) and a catalytic converter (13), which is mounted between the combustion engine (10) and the exhaust supercharger (14) **characterized** in that the catalytic converter (13) is a passive catalytic converter.

2. A combustion engine apparatus as claimed in claim 1, **characterized** in that the turbine of the exhaust supercharger (14) is a turbine operating by laminar flow, the said turbine operating by means of the laminar flow of the exhaust gases after the catalytic converter (13).

3. A combustion engine apparatus as claimed in claim 1 or 2, **characterized** in that the intake air of the combustion engine (10) is led to the inlet manifold (17) through the type of intercooler (16) which, together with the intake conduits (20), causes a pressure loss amounting to at most 200 mmH_2O.

4. A combustion engine apparatus as claimed in claim 1, 2 or 3, **characterized** in that the intercooler (16) is dimensioned so that the temperature of the supercharging air is 20-30°C higher than that of the ambient air after the intercooler.

5. A combustion engine apparatus as claimed in any of the claims 1-4, **characterized** in that the combustion engine (10) is a diesel engine, to the exhaust manifold (11) of which is connected a passive catalytic converter (13) through which exhaust gases are led to the exhaust supercharger (14) turbine operating by means of laminar flow, and that the intake air of the engine is led from the supercharger to the inlet manifold (17) of the engine through the type of intercooler (16) which, together with the intake conduits (20), causes a pressure loss amounting

to at most 200 mmH_2O, so that the temperature of the supercharging air is 20-30°C higher than that of ambient air after the intercooler.

6. A combustion engine apparatus as claimed in any of the claims 1-5, **characterized** in that a measuring element (40) is fixed to the exhaust manifold (11), and at the exhaust discharge opening of the exhaust supercharger (14) is fixed a measuring element (41) for measuring the temperatures of the exhaust gases and for simultaneously determining the degree of purification.

7. A combustion engine apparatus as claimed in any of the claims 1-6, **characterized** in that the catalytic converter (13) mounted between the combustion engine (10) and the exhaust supercharger (14) reaches, immediately after starting the engine (10), preferably at 0.4 bar supercharging pressure, the temperature required for exhaust gas purification.

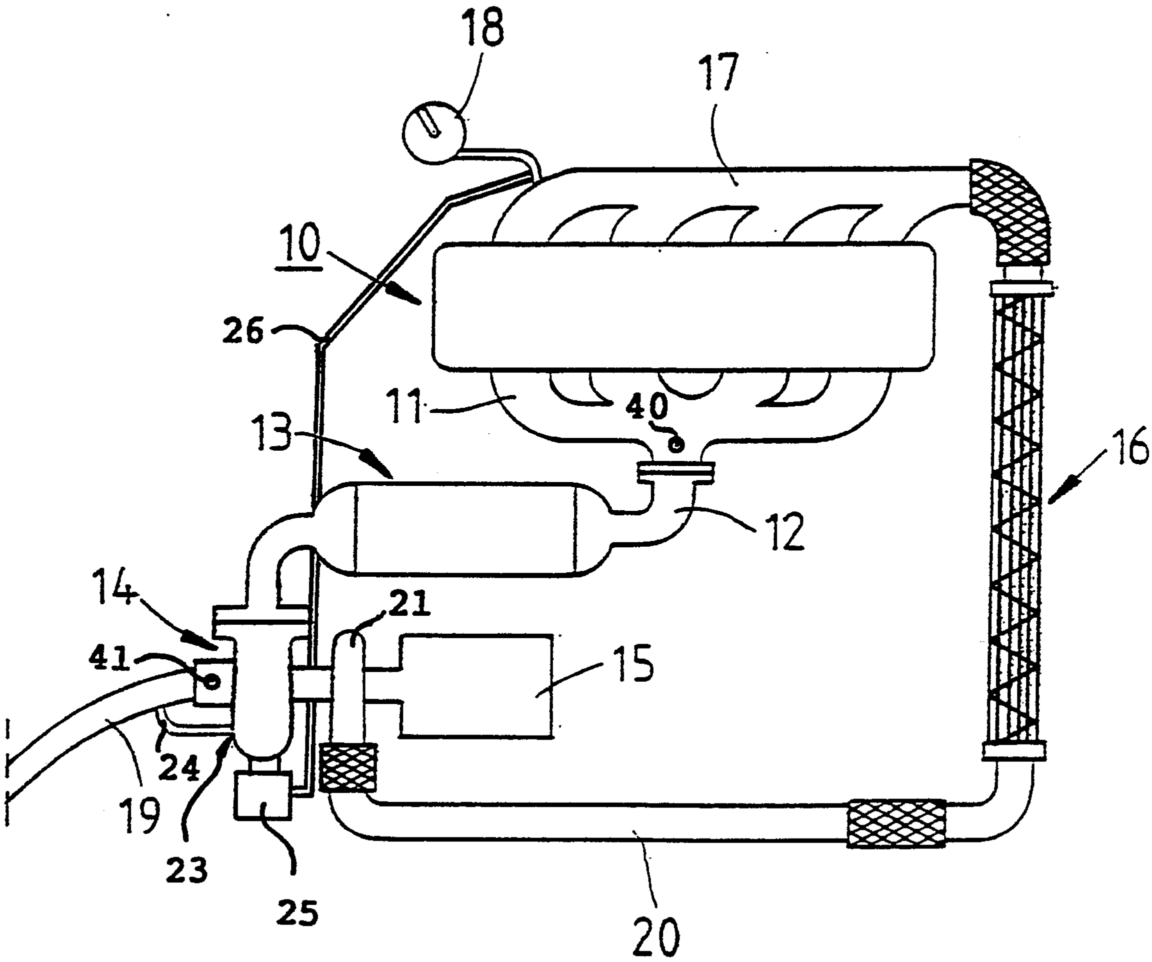

FIG. 1

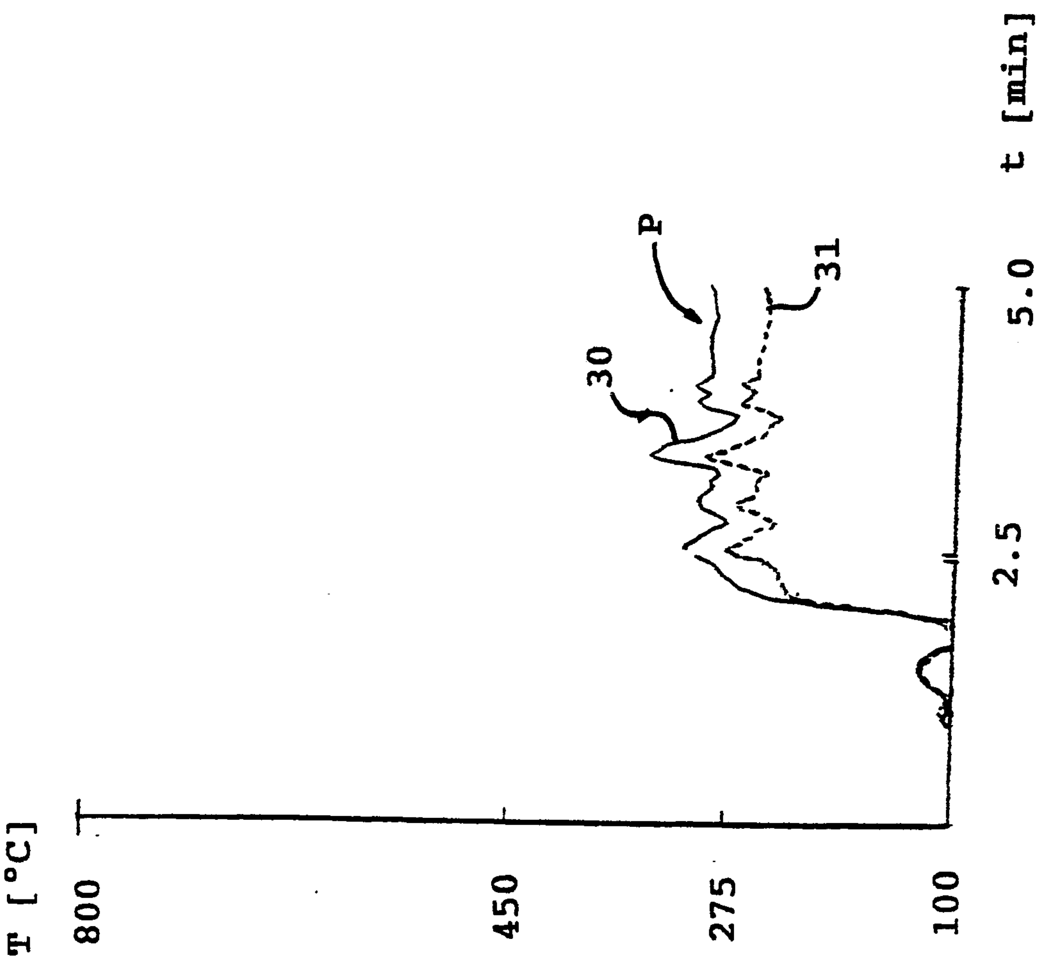

FIG. 2

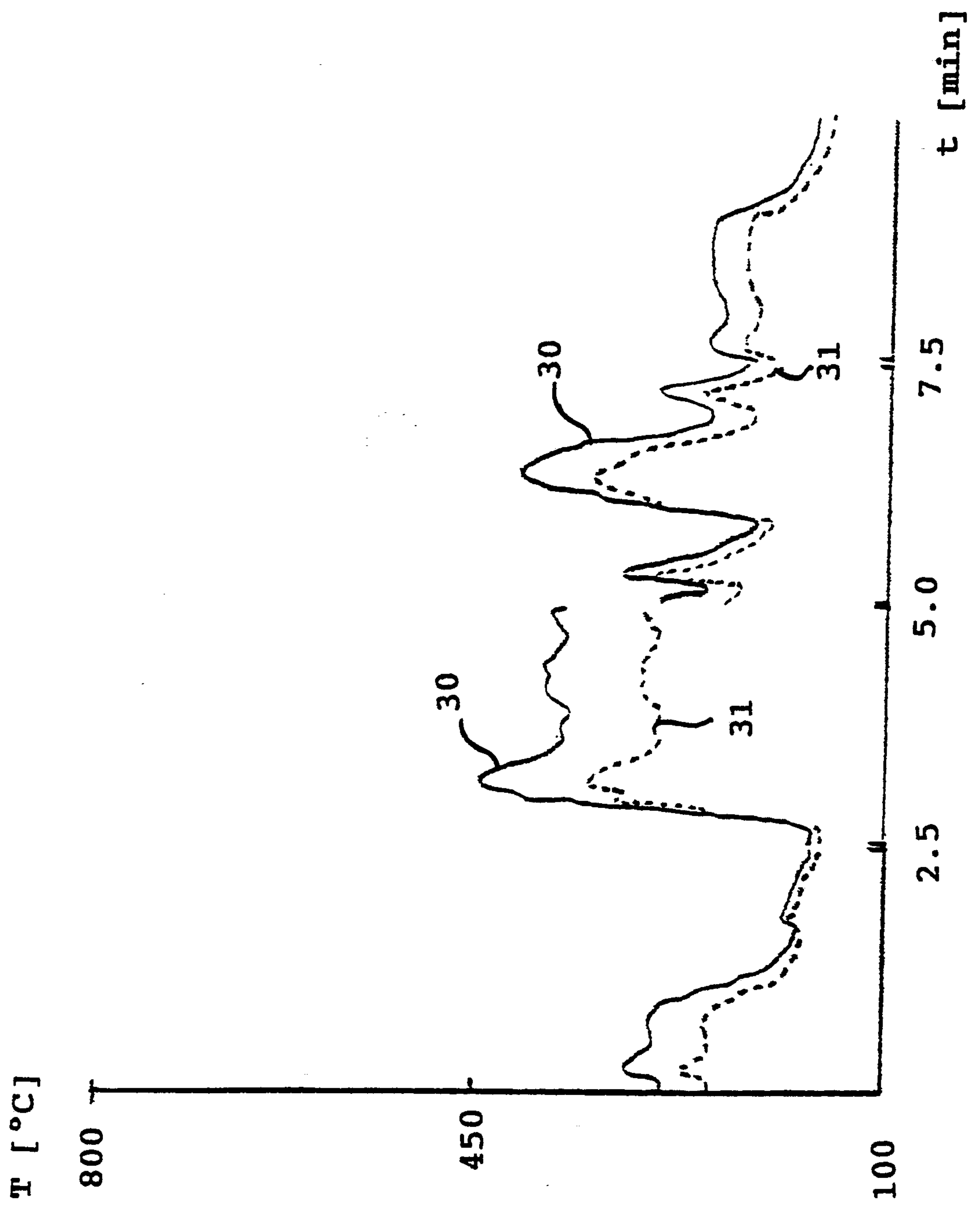

FIG. 3

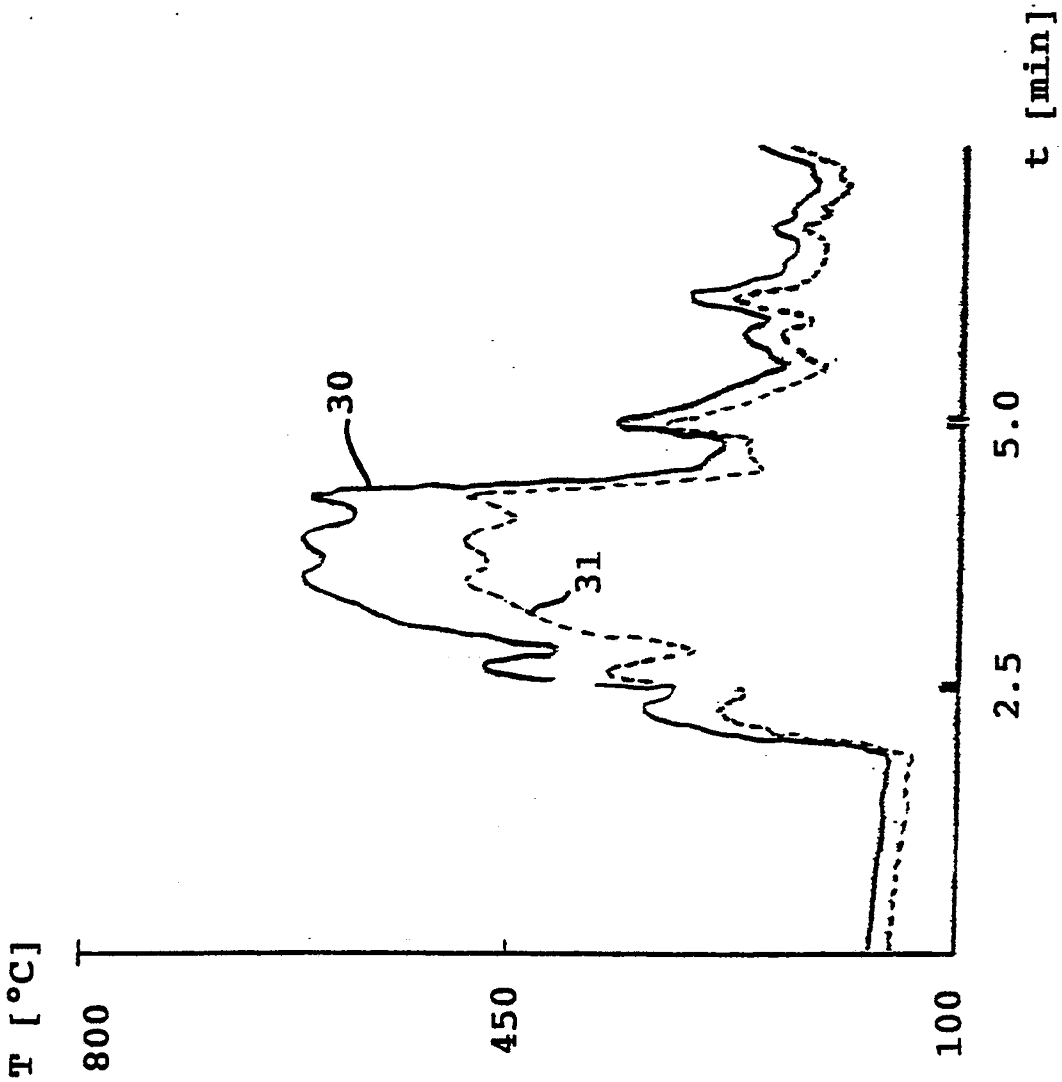

FIG. 4

INTERNATIONAL SEARCH REPORT

International Application No PCT/FI 92/00149

I. CLASSIFICATION OF SUBJECT MATTER (if several classification symbols apply, indicate all)[6]

According to International Patent Classification (IPC) or to both National Classification and IPC

IPC5: F 02 B 37/00

II. FIELDS SEARCHED

Minimum Documentation Searched[7]

Classification System	Classification Symbols
IPC5	F 02 B

Documentation Searched other than Minimum Documentation to the Extent that such Documents are Included in Fields Searched[8]

SE,DK,FI,NO classes as above

III. DOCUMENTS CONSIDERED TO BE RELEVANT[9]

Category *	Citation of Document,[11] with indication, where appropriate, of the relevant passages[12]	Relevant to Claim No.[13]
X	US, A, 4449370 (REAM) 22 May 1984, see the whole document --	1-7

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* Special categories of cited documents: [10]

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IV. CERTIFICATION

Date of the Actual Completion of the International Search	Date of Mailing of this International Search Report
31st August 1992	1992 -09- 02
International Searching Authority	**Signature of Authorized Officer**
SWEDISH PATENT OFFICE	Krister Bengtsson

ANNEX TO THE INTERNATIONAL SEARCH REPORT ON INTERNATIONAL PATENT APPLICATION NO. PCT/FI 92/00149

This annex lists the patent family members relating to the patent documents cited in the above-mentioned international search report.
The members are as contained in the Swedish Patent Office EDP file on 31/07/92
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Patent document cited in search report	Publication date	Patent family member(s)	Publication date
US-A- 4449370	84-05-22	NONE	